

Likhith Vardhan Goruputi

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Education

New York University

Master's, Computer Science

New York, NY

09/2025 - 05/2027

Teaching Assistant: AI for Cyber Security, Application Security

Vellore Institute of Technology

Bachelor of Technology in Computer Science (CGPA: 3.8/4)

Vellore, India

Sep 2021 – Aug 2025

Technical Skills

Programming Languages: Python, C++, C, CUDA, Go, Java, SQL

AI & ML: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, SciPy, Hugging Face, Transformers

GPU & Compilers: CUDA, CUDA Kernels, Tensor Cores, MLIR, LLVM, PTX, NVRTC

AI Tools: LangChain, LangGraph, OpenAI API, Google Vertex AI, Claude Code, Codex, Cursor, GitHub Copilot

Data, Cloud & Infrastructure: PostgreSQL, Redis, Kafka, AWS, Docker, Kubernetes, Jenkins, Git, REST APIs

Professional Experience

Roche

June 2026 – August 2026

Software Engineer Intern

Santa Clara, CA

- Worked on the **Digital Pathology Algorithms** team, migrating legacy **Java codebases to Python** and modernizing algorithm workflows for improved scalability.
- Built a **multi agent** AI framework to automate code migration and validation of 30+ pathology cases, reducing delivery time from **2 months to 2 weeks** with human in the loop validation .
- Automated algorithm deployments via **Jenkins cron** jobs and monitored production reliability with **Splunk**.

Quantum Geometric Intelligence (QGI) Lab

January 2025 – Present

Research Assistant

New York, NY

- Developing a computer vision pipeline using **YOLOv11, ByteTrack, OpenCV, and SoccerNet** to extract player, ball, and event-level data from broadcast soccer videos.
- Improved shot occurrence prediction AUC from **0.4517** → **0.7583** using **XGBoost + LightGBM** with off ball context across **696K+ actions and 426 matches**.

Quadrant Technologies

August 2023 – November 2023

Software Engineer Intern

Hyderabad, India

- Built a **domain specialized narcotics chatbot** using Django, React.js, LLaMA2 (LoRA fine tuned), delivering accurate, reliable responses to narcotics related inquiries
- Accelerated deployments by automating CI/CD pipelines for testing, linting, and container releases using Jenkins/Docker, publishing images to AWS ECR.

Academic Projects

BrandGuardian – ML Platform for Brand Safe Content Generation (PyTorch, Qwen2.5, DPO, LoRA, vLLM, Qdrant, Prefect)

- Built an end-to-end ML platform that generates, fact-checks, and routes marketing copy, cutting manual review from 100% to **23% of assets** at **99.1% precision** on unsafe content.
- Improved copy quality to a **71% win rate** over the SFT baseline in blind human evaluation by fine-tuning Qwen2.5-3B with **LoRA and DPO** on 4,800 preference pairs.
- Cut content validation cost **12×** with a Classifier cascade, while MLflow promotion gates blocked **6 regressions** before release.

LLM Inference Engine – GPU-Accelerated LLM Compiler (CUDA, C++, MLIR, LLVM, PyTorch)

- Built an **LLM inference engine** that compiles and executes optimized models on NVIDIA GPUs using a custom MLIR compiler pipeline and CUDA kernels.
- Reduced Llama-3.2-3B decode latency to **32.8 ms TPOT** using MLIR GPU lowering and kernel fusion, benchmarked against vLLM's **29.4 ms** on the same hardware.
- Reduced INT4 inference latency by **69%** using tensor core GEMM and in-register dequantization, achieving **8.1 ms TPOT** on an RTX 3060.

Half-Empty – LLM Sparsity & Quantization Engine (PyTorch, CUDA, Tensor Cores, LoRA, Transformers)

- Built an **LLM compression pipeline** that combines hardware-aware structured sparsity and quantization to accelerate model inference on NVIDIA GPUs.
- Reduced sparse-model perplexity degradation from **0.54 to 0.21** by recovering accuracy with Hessian-based pruning and knowledge distillation on Qwen2.5-1.5B.
- Achieved **2.34×** **faster inference** using Ampere 2:4 sparsity and INT8 quantization with layer-wise allocation on an RTX 3060.

Riptide – Distributed GPU Query Engine (CUDA, C++, NVRTC)

- Built a **distributed GPU query engine** that executes SQL analytics across GPU-accelerated workers using custom CUDA kernels.
- Reduced hash-join latency to **133 ms** using radix partitioning and shared-memory execution, processing **60M × 15M rows**.
- Reduced predicate execution time to **9.1 ms** by compiling query expressions into specialized CUDA kernels using NVRTC on 60M rows.

Achievements

- Winners of the **Eazo Global Hackathon 2026** conducted in **New York City**, recognized for developing an innovative AI-powered solution among participating teams.